

TsungWei Cheng Mike

Academic Curriculum Vitae

TsungWei Cheng (Mike) is a researcher specialising in digital fabrication, timber–earthen prototyping, and modular architectural systems. He holds a master's degree from NTUST and has completed academic exchanges at TU Berlin and TU Darmstadt.

He is currently a Research Collaborator at IBOIS–EPFL, developing bio-sourced architectural components that bridge low-tech materials with high-tech fabrication. He has published over ten papers on digital fabrication, architectural education, and modular systems.

His work focuses on transforming vernacular construction with bio-sourced materials into industrialised, digitally fabricated building methods, aiming to expand their architectural applications while reducing carbon emissions.

Professional Experience

Research Collaborator

Lausanne, Switzerland July 2026 - Present
Research Team: IBOIS, Prof. Yves Weinand
École Polytechnique Fédérale de Lausanne
Research focus: Bio-source materials and prototypes

Research Assistant

Taipei, Taiwan, Apr 2023 - June 2025
Research Team: GAS Lab, Prof. ShenGuan Shih
National Taiwan University of Technology and Science
Research focus: Dry Stacking System with Hierarchical Assembling by Rule-Based Approach and Interlocking Blocks Modular Design

Lecturer

Kaohsiung, Taiwan, Nov 2024 - June 2025
National University of Kaohsiung
Lecture Topics: Serious Gaming, innovative approach teaching students about BIM and MEP in architecture, Digital Fabrication with second hand materials

Internship

Yilan, Taiwan Jul 2021 - Jul 2024
Global Reengineering Co., Ltd.
Responsibilities: Optimisation of prefabricated concrete façade panels, structural skeleton visualisation, and design coordination.

Project Assistant

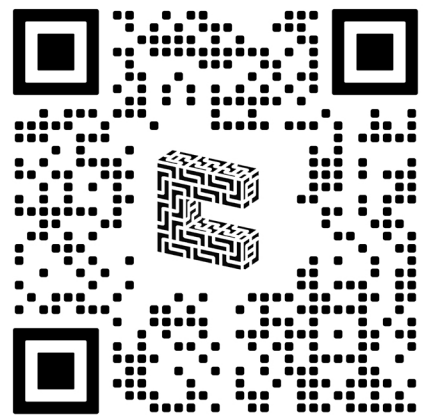
Taipei, Taiwan, Sep 2020 - Jul 2022
Research Team: GAS Lab, Prof ShenGuan Shih

Skill

Technical: Various Softwares (Rhino, Revit, 3D Max, AutoCad, Sketchup, Blender etc.), Programming (C#, C++, Javascript, Python, Typescript, R, etc), Git, Json, tool (Rasperry Pi), CNC Fabrication

Scientific working and Writing : LaTeX, Microsoft Office,

AI and Computational tool: Stable Diffusion, Chat GPT,



About

 mike861104@gmail.com

 [Tsung-Wei-Cheng](#)

 [宗偉-鄭-b76a45251](#)

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Lectures and Talks

Procedural Design for Interwoven Structures using SL Blocks: A computational Framework for Modular Architecture

ICCEA 2025 - Best Presentation Prize

Game-based Learning with Minecraft: Fostering Engagement in Mechanical-Electrical-Plumbing Systems Education

Asia-Pacific Conference on Education, Teaching & Technology 2025

Reducing Panel Complexity in Topological Interlocking Assemblies on Curved Surfaces

CADDRIA Conference 20th 2025

Lecturer at NUK

Teaching with Prof. YuPin Ma, National University of Kaohsiung

Nov 2023 - JUN 2025

Digital Fabrication

Minecraft Serious Gaming

Lego BIM Workshop

Rhino Advance Lecture

Design& Assembly: Symmetry Informed Assembly of Osteomorphic Block in Computer-Aided Design

The Research Exchange Day, Digital Design Unit at Technische Universität Darmstadt, Apr, 2024

Rule-Based Generation of Interwoven Assemblies in Architectural Design

CADDRIA Conference 19 th 2024

Grasshopper Lecture

Teaching with Prof. Fufei Yong,

National Taiwan University of Technology and Science, Nov 2023

Publications (Selected)

Tsung-Wei Cheng, Felicia Wagiri, Kevin Harsono, Ching-Yen Cheng, Shen-Guan Shih. (2025) Procedural Design for Interwoven Structures using SL Blocks: A computational Framework for Modular Architecture

Yu-Pin Ma, Tsung-Wei Cheng*, Rou Syuan Guo, Kevin Harsono, Felicia Wagiri, Shen-Guan Shih (2025). Game-based Learning with Minecraft: Fostering Engagement in Mechanical-Electrical-Plumbing Systems Education

Kevin Harsono, Tsung-Wei Cheng*, Felicia Wagiri, Ye Yint Aung, Shen-Guan Shih (2025) Modular and mortarless masonry: design, fabrication, and reusability assessment framework for topological interlocking assembly

Tsung-Wei Cheng, Kevin Harsono, Yuxi Lui, Ching-Yen Chen, Shen-Guan Shih, Oliver Tessmann. (2024). Beyond Babel: Towering With Minimal Communication. *Proceedings of the IASS 2024 Symposium*. International Association for Shell and Spatial Structures (IASS 2024). Switzerland.

Felicia Wagiri, Shen-Guan Shih, Yu-Chuan Kao, Tsung-Wei Cheng, Mu-Kuan Lu. (2024). Bending-active molds for pre-fabricated concrete shells. *Proceedings of the IASS 2024 Symposium*. Redefining the Art of Structural Design (IASS 2024). Switzerland.

Yuxi Liu, Boris Belousov, Tim Schneidern, Kevin Harsono, Tsung-Wei Cheng, Shen-Guan Shih, Oliver Tessman (2024). Advancing Sustainable Construction: Discrete Modular Systems & Robotic Assembly. *Sustainability* 16(15):6678. <https://doi.org/10.3390/su16156678>

Kevin Harsono, Shen-Guan Shih, Ye Yint Aung, Felicia Wagiri, Tsung-Wei Cheng (2024). RULE-BASED GENERATION OF INTERWOVEN ASSEMBLIES IN ARCHITECTURAL DESIGN: A COMPUTATIONAL APPROACH INTEGRATING ATTRIBUTE GRAMMAR. *Proceedings of the 29th International Conference on Computer-Aided Architectural Design Research in Asia (CAADRIA 2024)*. At: Singapore.

Cheng, T.W., Wu, B.R., Chiang, Y.T., Liao, C.T. (2018). The Study on Unlift of Exterior Tile in Nanzih District of Kaohsiung City Processings of the 30th Annual Conference of the Architecture Institute of Taiwan

Tsung-Wei Cheng. (2022), Production Rules for the Arrangement of Osteomorphic Block.

Education

Exchange Researcher

Darmstadt, Germany, April 2024

Auto-Replicating Robotic Assembly of SL Blocks for the Hierarchical and Reconfigurable Construction

Project headed by Prof. ShenGuan Shih, and Prof Oliver Tessmann

Exchange Student

Berlin, Germany, Sep 2022-Mar 2023

Circular Economy and Ontology of Building Information Modelling

Project headed by Prof. Timo Hartmann

the Department of Civil Engineering, Technische Universität Berlin

Master of Architecture

Taipei, Taiwan, Sep 2020-Jul 2023

Production Rule for the design patterns of modular design system

Thesis: Production Rules for the Arrangement of Osteomorphic Block

Supervision headed by Prof. ShenGuan Shih

GAS Lab, National Taiwan University of Technology and Science

Bachelor of Creative Design and Architecture

Taipei Taiwan, Sep 2016-Jul 2020

Colledge of Humanities and Social Sciences

National University of Kaohsiung

Funding

DAAD-PPP_Program (2nd)- Auto-Replicating Robotic Assembly for SL Blocks for the Hierarchical and Reconfigurable Construction

Jan 2025 - Jun 2025

National Science and Technology Council

STEAM Course

Jan 2025 - Jun 2025

National University Kaohsiung

DAAD-PPP_Program (1st)- Auto-Replicating Robotic Assembly for SL Blocks for the Hierarchical and Reconfigurable Construction

Jan 2024 - Dec 2025

National Science and Technology Council

Digital Application for Sustainable Built Environment Program

Sep 2023 - Mar 2024

National Taiwan University of Technology and Science

Exploring the Use of Truchet Tiling in Architectural Facade Design

Sep 2022 - Mar 2023

Scholarship of Chung Hwa Rotary Education Foundation